

Evaluation Metrics — Design & Results

Expanded comparison panel for synthetic EV charging generators (combined dataset, 140 users / 28,159 sessions).
Implemented in `benchmarking/metrics.py`.

1. Why we expanded beyond KL

A single population-level KL on the marginals rewards a generator for matching the *aggregate* shape of arrival, duration and energy. On that metric our method (0.008) only narrowly beats the neural GMMNet baseline (0.012). But marginal KL is blind to three things that matter for a charging model: whether each *individual user* is reproduced, whether the *dependence* between variables is preserved, and whether the *sequential* structure (the idle gaps between a user's sessions) is captured. The panel below measures all of these so the comparison reflects what the model is actually for.

2. What each metric measures

Population marginal KL

pop_KL. Mean over the three variables of $KL(p_{\text{real}} \parallel p_{\text{synth}})$ on Laplace-smoothed histograms. Aggregate marginal fidelity. *Lower is better.*

Per-user KL

perUser_KL. For every real user u , the KL between that user's marginal and the generator's output for u , averaged over users (weighted by sessions/user). Our method has a per-user profile; the pooled baselines have no user identity, so each user is scored against the single pooled output. This is the direct test of *heterogeneity*. *Lower is better.*

Dependence: Kendall-tau error

tau_err. $|\tau_{\text{real}} - \tau_{\text{synth}}|$ averaged over the three variable pairs, where τ is Kendall's rank correlation. Measures whether the cross-variable *association* is preserved (e.g. late arrival $\rightarrow$ shorter duration). *Lower is better.*

Dependence: tail-dependence MAE

tail_MAE. Error in the empirical upper/lower tail-dependence coefficients $\lambda = P(V \text{ extreme} \mid U \text{ extreme})$ at the 5%/95% quantiles, averaged over pairs and both tails. Captures joint *extremes* (simultaneous long-duration & high-energy events). *Lower is better.*

Distinguishability (C2ST)

C2ST_AUC. A gradient-boosted classifier is trained to separate real from synthetic on (arrival, duration, energy); we report cross-validated ROC-AUC. **0.5 = indistinguishable (ideal)**; 1.0 = trivially separable. A single holistic score of joint realism.

Grid utility (load profile)

load_MAE. Each session's energy is spread at constant power over its plug-in window to build the average daily load curve (kWh/h per session); we report the mean absolute error vs. the real curve. This is the quantity a distribution-grid study cares about. *Lower is better.*

Sequential structure (idle gap)

gap_KL. KL between the real and synthetic distributions of the idle gap between a user's consecutive sessions. Only a sequential model emits a gap; the pooled baselines generate independent sessions, so this is **N/A** for them by construction. *Lower is better.*

Privacy / memorisation (DCR)

DCR. Median distance from each synthetic record to its nearest real record (standardised). Compared against a real-vs-real reference (0.043 here); a similar value means the generator is not copying real sessions. Near-zero would signal a privacy leak.

3. Results (combined dataset)

Metric	Ours (SMC)	GMMNet	EV-SDG	Best
pop_KL ↓	0.0079	0.0121	1.5330	Ours

perUser_KL ↓	0.5103	0.5101	2.0310	tie (Ours/GMMNet)
tau_err ↓	0.0806	0.0347	0.0947	GMMNet
tail_MAE ↓	0.0875	0.0498	0.0870	GMMNet
C2ST_AUC →0.5	0.642	0.590	0.741	GMMNet
load_MAE ↓	0.0939	0.0183	0.1188	GMMNet
gap_KL ↓	0.0362	N/A	N/A	Ours (only)
DCR (ref 0.043)	0.050	0.041	0.045	all ≈ ref

4. Honest interpretation

- **Our clear, structural win is the idle gap (gap_KL 0.036).** The pooled baselines cannot be scored on it at all — they generate independent sessions with no notion of a user's session sequence. This is the semi-Markov contribution and it is unique to our method.
- **We also win on marginal KL** (0.008 vs 0.012 vs 1.53).
- **GMMNet is genuinely strong and beats us on the static joint.** It preserves cross-variable dependence (tau_err, tail_MAE), is harder to distinguish from real (C2ST 0.590 vs our 0.642), and reproduces the aggregate load curve better (load_MAE 0.018 vs 0.094). This is expected: GMMNet explicitly learns the joint density, whereas our framework models per-bin marginals plus transitions, so it trades some joint-dependence fidelity for interpretability and sequential structure. Reporting this straight is more credible than hiding it.
- **Per-user KL is currently a tie** and is dominated by finite-sample noise (each user has few sessions, so even a perfect model incurs ~0.5 KL). It does not yet isolate the heterogeneity advantage — a moment-based per-user metric (per-user mean/quantile error) would be cleaner; flagged as a refinement.
- **EV-SDG is clearly the weakest** baseline on every axis.
- **No memorisation:** all DCR values sit at the real-vs-real reference (~0.04–0.05).

Implication for the paper: lead the contribution with the *sequential / per-user semi-Markov* structure (idle gaps, session ordering) that competing generators structurally cannot produce, rather than claiming dominance on static marginals or joint dependence — where a strong neural baseline (GMMNet) is competitive or better.